

Lec 03: Performance modeling of computer networks

Computer Networks (CSE232) Section A

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These slides are inspired and adapted from:

1. *Computer Networking: A Top-Down Approach*
6th edition, Jim Kurose, Keith Ross, Pearson, 2017



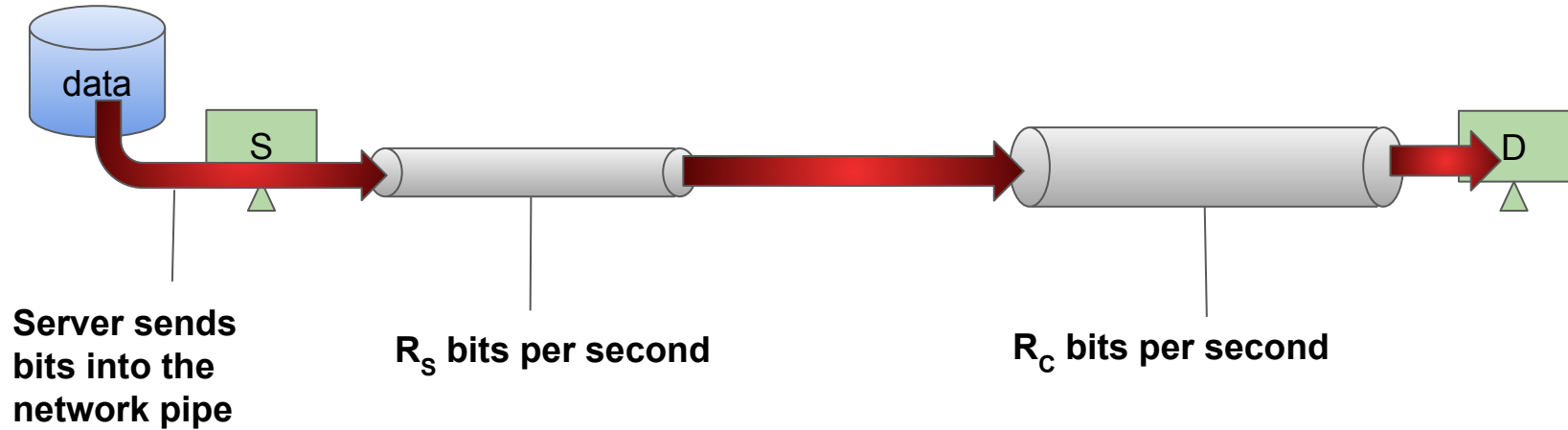
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Modeling the performance of computer networks

Two main attributes to measure network performance

- **Throughput**
 - Rate at which bits are transferred between the sender and receiver
- **Delay**
 - Time taken for a bit to reach from the sender to receiver

Throughput



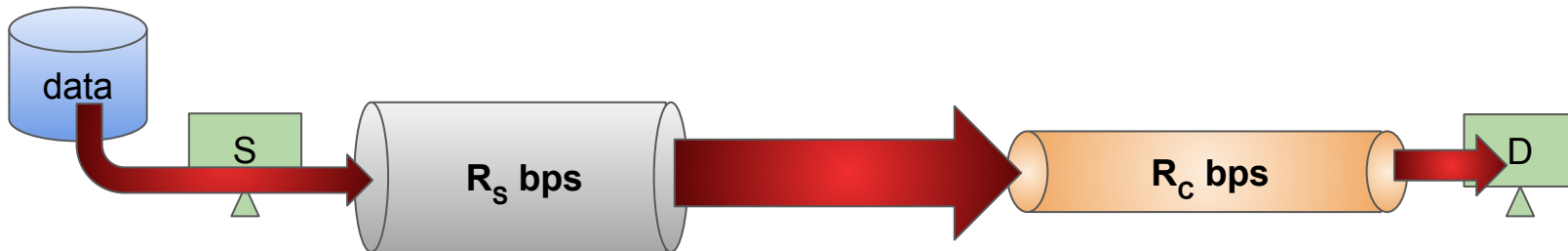
Throughput: Rate at which bits are transferred between the sender and receiver (bits per second)

Throughput (contd.)

$R_s < R_c$ What is the average end-end throughput?

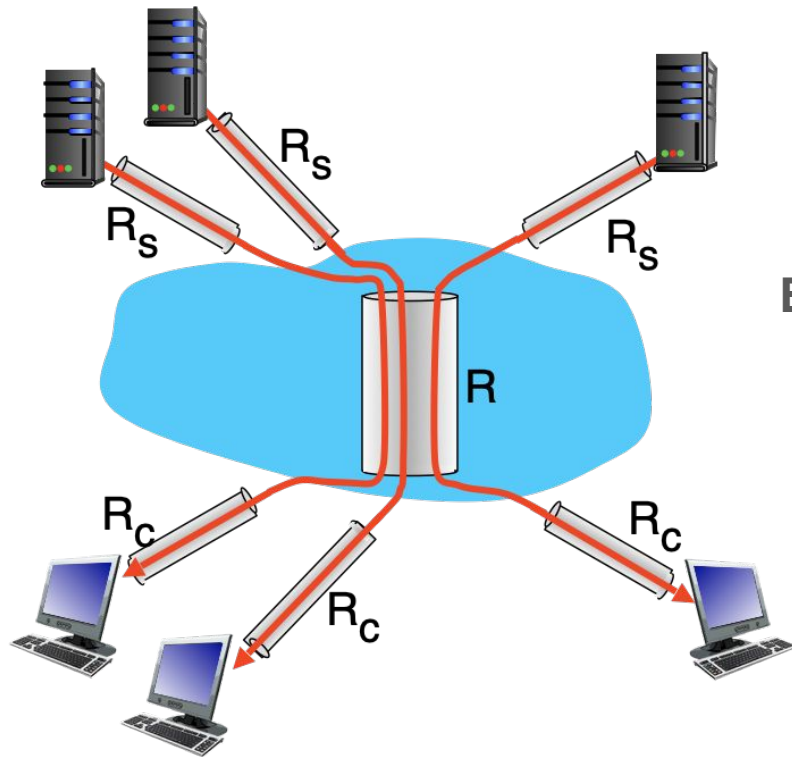


$R_s > R_c$ What is the average end-end throughput?



Bottleneck link on the path constraints end-end throughput

Throughput: Internet scenario



10 connections (fairly) share
backbone bottleneck link R bits/sec

End-to-end throughput per connection?

- $\min(R_S, R_C, R/10)$
- Typical bottleneck, R_C or R_S